SCADA Design Manual - Rev. 2.5

CONFIDENTIAL INFORMATION - SHARED UNDER NDA ONLY

PRODUCT SPECIFICATIONS

All specifications and descriptions contained in this document are verified to be accurate at the time of printing. However, because continuous improvement is a goal at Tesla, we reserve the right to make product or documentation modifications at any time, with or without notice.

The images provided in this document are for demonstration purposes only. Depending on product version and market region, details may appear slightly different.

This document does not create contractual obligations for Tesla or its affiliates, except to the extent expressly agreed in a contract.

LATEST REVISIONS

Documents are periodically updated. To ensure you have the latest revision of this document, visit the Tesla Partner Portal at *https://partners.tesla.com/*.

ERRORS OR OMISSIONS

To communicate any inaccuracies or omissions in this document, contact us via email at *energy-pubs@tesla.com*.

Reference Documents and Terminology

Reference Documents

- Tesla Energy Controls and Communications Manual - *https://partners.tesla.com/home/en-US/content/download/Tesla_Energy_Controls_and_Communications_Manual.pdf*
- SCADA Architecture Diagrams - *https://partners.tesla.com/home/en-US/content/download/SCADA_Architecture_Diagrams.pdf*
- Tesla Industrial Energy Approved Vendors List - *https://partners.tesla.com/home/en-US/content/download/Tesla_IndustrialEnergy_Approved_Vendor_List.pdf*
- Megapack Deployment Overview - *https://partners.tesla.com/home/en-us/content/download/megapack_deployment_overview.pdf*
- Megapack Deployment Checklists - *https://partners.tesla.com/home/en-us/content/download/megapack_deployment_checklists.zip*
- Megapack 2 Design and Installation Manual - *https://partners.tesla.com/home/en-US/content/download/Megapack_2_Design_and_Installation_Manual.pdf*
- Megapack 2 XL Design and Installation Manual - *https://partners.tesla.com/home/en-US/content/download/Megapack_2_XL_Design_and_Installation_Manual.pdf*
- Megapack Site Design Manual - *https://partners.tesla.com/home/en-US/content/download/Megapack_Site_Design_Manual.pdf*
- Powerpack 2 System Site Design Manual - *https://partners.tesla.com/home/en-US/content/download/Powerpack_2_System_Site_Design_Manual.pdf*
- Powerhub User Manual - *https://partners.tesla.com/home/en-US/content/download/Powerhub_User_Manual.pdf*
- Powerhub API Manual - *https://partners.tesla.com/home/en-US/content/download/Powerhub_API_Manual.pdf*
- Configuration Specification: Using I/O Devices with Tesla Products - *https://partners.tesla.com/home/en-US/content/download/Configuration_Specification_Using_IO_Devices_with_Tesla_Products.pdf*

Acronyms and Terminology

Common SCADA-related terms and acronyms as they may be used in this document are listed below.

Table 1. Abbreviations, Acronyms, and Definitions

Acronym / Term	Definition
AC	Alternating current
BESS	Battery energy storage system
BOM	Bill of material
BOP	Balance of plant
DHCP	Dynamic host configuration protocol
FNE	Field Network Enclosure
GUI	Graphical user interface
IO; RIO	Input/output; remote input/output

Acronym / Term	Definition
ISO	Independent System Operator
ISP	Internet service provider
LAN	Local area network
MV	Medium voltage
OEM	Original Equipment Manufacturer
POI	Point of Interconnection
Powerhub API	Powerhub application programming interface
PV	Photovoltaic
RIG	Remote Intelligent Gateway (used as a telemetry RTU)
RTAC	Real-Time Automation Controller (an SEL product/term)
RTU	Remote terminal unit
SCADA	Supervisory control and data acquisition
SFP	Small form-factor pluggable
Subnet	Logical subdivision of an IP network
TCP	Transmission control protocol
TSC	Tesla Site Controller
UI	User interface
UPS	Uninterruptible power supply
VLAN	Virtual local area network
WAN	Wide area network

1 Introduction and Scope

This document outlines the design principles of a SCADA system for Tesla's battery energy storage systems (BESS), Tesla-supported photovoltaic (PV) systems, Tesla Site Controller, and other relevant products and offerings. It is intended to support SCADA engineers and other relevant stakeholders in designing an end-to-end SCADA solution with Tesla products. This document provides Tesla's OEM SCADA design and equipment requirements as well as, for reference, Tesla's internal EPC standards and consequent recommendations.

To best guide the various target audiences of this document, the main sections are grouped as below:

- *SCADA Package and Tesla's Offerings* on page 5
- *Network Architecture* on page 8
- *Selecting SCADA Hardware* on page 21

If you have any questions about this document or the SCADA offering, please contact your Tesla representative.

Table 2. Standard Division of Responsibilities

Phase	Tesla	EPC
Design		
Definition of Tesla's standard network architecture, networking requirements, approved equipment, and provided-equipment specifications	Responsible	
SCADA system and communication network architecture and details (physical & logical)	Must Inform	Responsible
Integration of Tesla's requirements (physical & logical)	Must Inform	Responsible
Align customer expectations with Tesla's requirements and offerings	Must Consult	Responsible
Tesla's Remote Connection to the project site	Must Inform	Responsible
Materials Specification and Supply		
Tesla Site Controller	Responsible	Must Inform
Communications network (routers, switches, fiber, copper, wireless, etc)	Must Consult	Responsible
SCADA equipment and enclosures (meters, IO devices, jump servers, data servers, RTAC, firewalls, HMI, FNEs, POI SCADA, etc)	Must Inform	Responsible
Construction, Commissioning, Operations, Maintenance		
Communications network	Must Consult	Responsible
SCADA equipment and enclosures	Must Consult	Responsible
Commissioning (Program/Configure, Test, Calibrate)		
The communication network (routers, modems etc)	Mus tInform	Responsible
SCADA Enclosures (Standard Tesla Site Controller Enclosure, FNE, POI SCADA)	Must Inform	Responsible

2 SCADA Package and Tesla's Offerings

A SCADA system is a collection of computer systems with various logical and physical interfaces that monitor and control a facility's operation.

This section summarizes Tesla's SCADA offerings for control, data acquisition (telemetry), data storage (Historian), and data access:

- *Controls* on page 5
- *Telemetry* on page 6
- *Historical Data Storage (Historian)* on page 6
- *Graphical User Interface* on page 7

Talk to your Tesla sales team regarding pricing for Tesla's various SCADA offerings.

2.1 Controls

The customer is responsible for determining and implementing the system's control scheme.

Tesla offers solutions to meet a variety of control requirements, as described below.

2.1.1 Standard Control Strategies

The Tesla Site Controller by default has some pre-programmed modes of operation for the customer's use as described in the *Tesla Energy Controls and Communications Manual* including:

- Direct Real and/or Reactive Power Mode
- Site Control Mode; Scheduler, Go To Energy, Non-Export
- Frequency & Voltage Support
- Islanding Modes of Operation

Customers often have additional SCADA equipment that manages custom logic and interfaces with the Tesla Site Controller to dispatch the standard control strategies.

Additional monitoring and control functionality is available via Tesla-approved I/O devices. For details, see *I/O (Input/Output) Devices* on page 32.

2.1.2 Control via Powerhub

See *Graphical User Interface* on page 7 and the *Powerhub User Manual* for more information.

2.1.3 Autonomous Control Software

Autonomous control offerings from Tesla include:

- Tesla Opticaster
- Tesla Autobidder
- Tesla Microgrid Controller

2.1.4 Custom Control Strategies

Custom SCADA control solutions include:

- Multiple control sources - Arbitrage of control based on a pre-defined priority between multiple sources. This may include arbitration between the utility/ customer and the region's ISO, or arbitration of parallel local and remote commands to the Tesla Site Controller from Powerhub Local and a customer's controller/ RTU.
- Miscellaneous control logic - For example: transformer loading management, utility network-based charging schedules.
- Custom protection control scheme - Schemes to execute certain controls based on protection triggers; for example, tripping certain circuit breakers based on transformer over temperature or fire detection alerts.

2.2 Telemetry

Telemetry refers to non-graphical real-time and historical data collected at a project site and transmitted directly to a customer's device (RTU and/or server) for monitoring and storage.

The customer is responsible for creating a project-specific data points list. The data points list should define communication protocol, tag/point names, data types, address/indices, source, scaling, and units. Tesla can provide a generic template upon request.

Where Tesla has an operations or maintenance obligation such as an availability guarantee that spans step-up transformers, Tesla requires that SCADA points for transformer monitoring be gathered and exposed to the Tesla Site Controller or Tesla-managed RTAC via Modbus TCP.

Where an AHJ requires a third-party smoke, gas, or fire detection and/or suppression system, the associated SCADA points should be exposed to the Tesla Site Controller or a Tesla-managed RTAC via Modbus TCP.

Three groups of data can be made available by Tesla, as outlined below:

2.2.1 Standard Local Telemetry (Tesla Site Controller)

The Tesla Site Controller offers data points interface in the format and order listed in the *Tesla Energy Controls and Communications Manual*.

2.2.2 Custom Local Telemetry

When contracted, Tesla can design custom local telemetry solutions with an RTAC as described in *SEL RTAC* on page 23.

2.2.3 Remote API-Based Telemetry

The customer can use Tesla's cloud-based Powerhub API for remote telemetry. For the available points and other information, see the *Powerhub API Manual*.

2.3 Historical Data Storage (Historian)

Tesla offers two options for historical data storage: Remote Historian and Local Historian:

2.3.1 Remote Historian

Customers can leverage Tesla's existing databases that store data for their sites. Data can be viewed and retrieved in different ways:

- Powerhub Cloud provides remote visualization of historical performance data of the on-site assets to explore detailed data channels and track generation and consumption over time. Powerhub's Graph View provides historical data for pre-defined groups of telemetry data, which can be viewed over configurable time frames. For more information see the *Powerhub User Manual.*
- Data can be downloaded to .CSV files from Graph View. For more information, see the *Powerhub User Manual.*
- Historical data can also be accessed through Tesla's Powerhub API. For more information see the *Powerhub API Manual.*

2.3.2 Local Historian

If onsite storage and access of data is required, Tesla can be contracted to provide a locally hosted third-party historian. Typically, Tesla will utilize *Wonderware/AVEVA's* local Historian to perform Historian functions such as:

- Making data available for viewing and downloading, with features such as 15-minute average data download and ODBC connection to Microsoft Excel and SQL.
- Allowing configuration of up to 500 tags with a granularity of one second.
- Allowing access to data through a REST API.
- Providing data in .CSV file format.

2.4 Graphical User Interface

Projects often require a graphical user interface (GUI) for access to their telemetry and controls. Tesla GUI options include:

2.4.1 Powerhub

Powerhub is Tesla's GUI software, offering telemetry, controls, and Historian options in two different packages:

Powerhub Cloud: Powerhub's cloud-based GUI, Powerhub Cloud, is hosted on Tesla servers and accessible via a web browser. Powerhub Cloud allows monitoring of all available sites in customer's portfolio.

Powerhub Local: The Powerhub Local package provides a locally available GUI hosted on the Tesla Site Controller. The local GUI can provide both monitoring and control of a single site.

For a full description and delineation of the packages, refer to the *Powerhub User Manual.*

2.4.2 Wonderware

When contracted, Tesla can provide a custom GUI. Typically, Tesla will utilize *Wonderware/AVEVA's* software platform.

3 Network Architecture

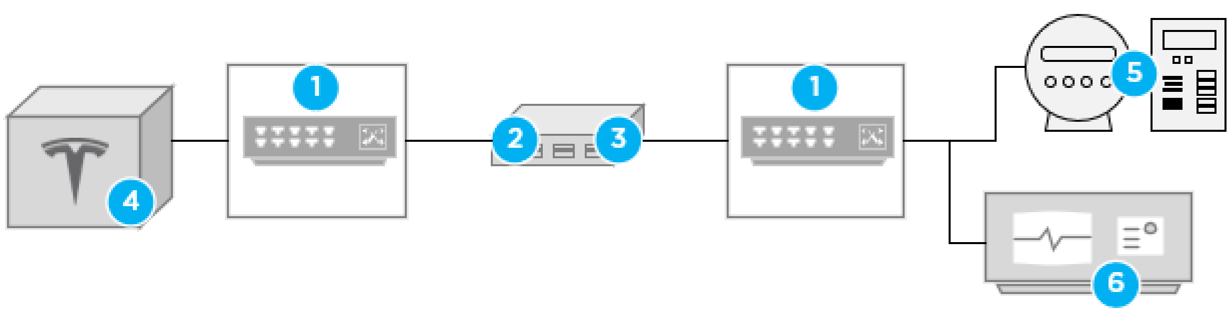
This section provides requirements and guidelines for designing the networking and communication architecture for a Tesla industrial energy project.

Tesla requires network separation between the Tesla Network (*Tesla Network* on page 9) and the Customer Network (*Customer Network* on page 10). The two networks are described further below. Network separation facilitates the Tesla Site Controller's function as a DHCP server to the battery system (BESS) units, facilitates network performance by isolating and prioritizing BESS traffic, and increases network security by restricting access to the BESS traffic. There are two options for satisfying Tesla's network separation requirement:

- Physical separation via independent network switches as shown in *Example of Physical Network Separation by Network Switches* on page 8.
- Logical separation with VLANs, a conceptual example of which is shown in *Example of Logical Network Separation by VLANs* on page 9.

Tesla may require the use of specific network switches as discussed in *Network Switches* on page 25.

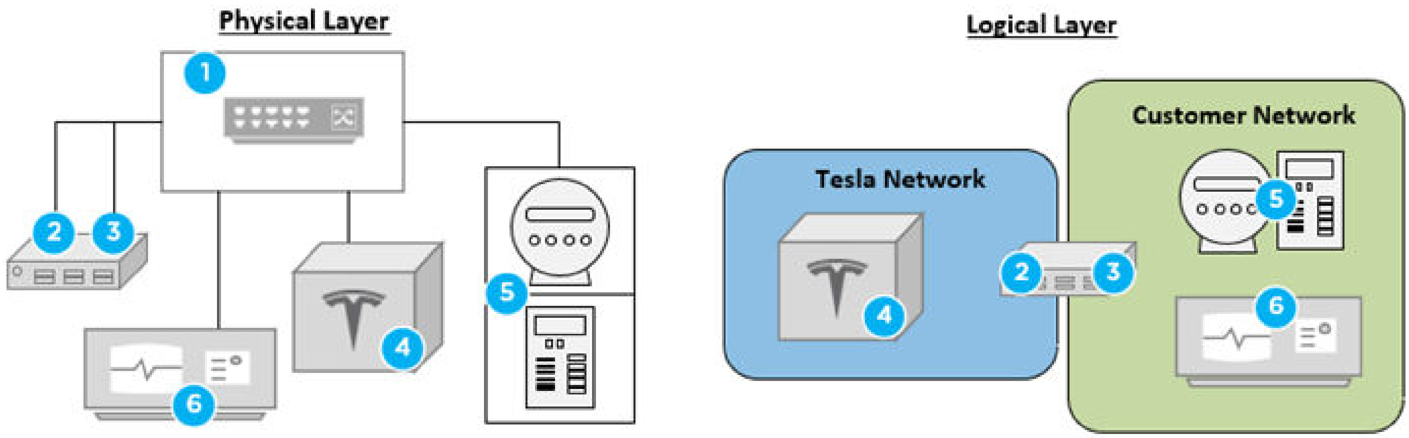
Figure 1. Example of Physical Network Separation by Network Switches



1. Network switch and/or routing device
2. Tesla Site Controller LAN 2 RJ45/Ethernet Port
3. Tesla Site Controller LAN 1 RJ45/Ethernet Port
4. Tesla battery system – for example, Megapack or Powerpack
5. Balance Of Plant Equipment, such as meters, protection relays, and I/O devices
6. Customer / third-party SCADA equipment

See *SCADA Architecture Diagrams* on the Partner Portal for detailed diagrams of physical network separation.

Figure 2. Example of Logical Network Separation by VLANs



1. Network switch and/or routing device
2. Tesla Site Controller LAN 2 RJ45/Ethernet Port
3. Tesla Site Controller LAN 1 RJ45/Ethernet Port
4. Tesla battery system – for example, Megapack or Powerpack
5. Balance Of Plant Equipment, such as meters, protection relays, and I/O devices
6. Customer / third-party SCADA equipment

3.1 Networks and VLANs

3.1.1 Tesla Network

The Tesla Network contains, at a minimum, the Tesla Site Controller (LAN 2 Ethernet port) and Tesla battery system inverters.

Where physically isolated from the Customer Network, any network monitoring, network access, and data transfer to/from the Tesla Network must be performed by a Tesla controlled multi-homing device. The Tesla Site Controller exposes a limited Tesla Network data set (for example, meter data), for more information see the *Tesla Energy Controls and Communications Manual*. If that data set is insufficient, Tesla can be contracted to program a multi-homed SEL RTAC. For more information, see *SEL RTAC* on page 23.

Where logically isolated from the Customer Network, the customer is responsible for the routing of data between VLANs.

To maintain site-to-site consistency and ease of troubleshooting, Tesla requires that the Tesla Network follows the standard IP addressing scheme in *Appendix A: Standard IP Address Range* on page 34.

3.1.1.1 Tesla Network Bandwidth Requirements

Tesla requires that the Tesla Network is designed around a data bandwidth of at least 1 (one) Mbps per Tesla BESS unit per active Tesla Site Controller (required network bandwidth = 1 Mbps * # TSCs * # BESS Units). In this case a BESS unit refers to an entire Megapack or Powerpack. This design bandwidth accounts for traffic from the BESS unit and possible BOP equipment including meters, IO devices, and a Tesla managed RTAC (when used). For example, a site with redundant (two) Tesla Site Controllers can accommodate up to 500 BESS units where the Tesla Network backbone is designed with a 1 Gbps bandwidth.

Tesla has standardized its SCADA equipment requirements, specifications, and offerings around a Tesla Network backbone of 1 (one) Gbps. For example, the Tesla-designed field network enclosures utilize 1 Gbps fiber transceivers to create the fiber ring loops described in *Field Network Enclosure* on page 30.Tesla may not support troubleshooting of SCADA networks that deviate from these standards.

3.1.2 Customer Network

The Customer Network contains, at a minimum, the Tesla Site Controller (LAN 1 Ethernet port).

System operators can interface with the Tesla Site Controller over the Customer Network using TCP over Modbus, DNP3, or REST as described in the *Tesla Energy Communications and Controls Manual.*

By default, the Tesla Site Controller's LAN 1 Ethernet port is configured as a DHCP client and accepts an IP address provided by a customer DHCP server on startup. The Tesla Site Controller's LAN 1 Ethernet port can also be configured with a static IP address as described in *Appendix C: Network Configuration of the Tesla Site Controller's LAN 1 Ethernet Port* on page 38.

3.1.2.1 Customer Network Bandwidth Requirements

At a minimum, Tesla requires a 100 Mbps network connection when the Tesla Remote Connection is provided over the Customer Network.

3.1.3 Devices and Network Placement

For device-specific guidance, see *Selecting SCADA Hardware* on page 21.

3.2 Tesla Remote Connection

Tesla requires two types of remote connections for warranty and maintenance of Tesla industrial energy storage systems:

- Data logging, which periodically collects operational data from the system for maintenance and warranty purposes.
- Remote access, which is required by the Tesla service teams to troubleshoot and diagnose issues, control the system for testing, and deliver firmware updates.

Tesla acknowledges that some customers have policies that do not allow third parties to have direct and persistent inbound/outbound connections to the assets owned and operated by the customer. This section briefly describes Tesla's standard remote connection options and alternative remote connection options to comply with customer requirements. The alternative remote connection methods have an associated additional price to cover for deployment and operation and maintenance cost. Contact your sales representative for more details on cost.

3.2.1 Standard Remote Connections

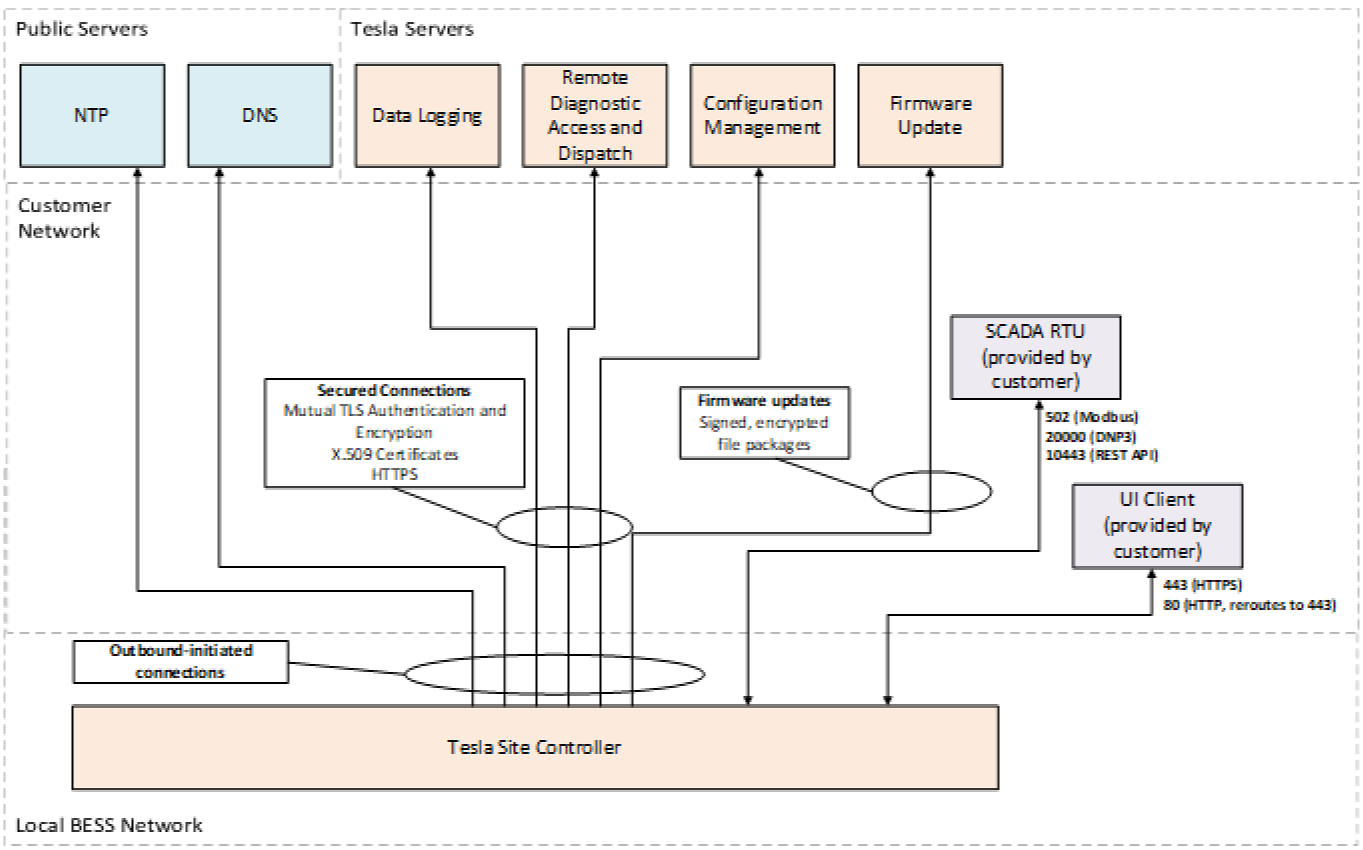
The Tesla Site Controller requires access to the Tesla servers either via the integrated cellular modem or an unrestricted hardline internet connection provided by the customer over the Customer Network and the Tesla Site Controller's LAN 1 Ethernet port.

Tesla enforces high cybersecurity standards to meet the needs of Tesla energy products. The Tesla Site Controller has an integrated firewall. For both standard remote connections (cellular and LAN 1), the Tesla Site Controller establishes outbound-initiated connections with public NTP servers for time synchronization and with public DNS servers for hostname resolution. The Tesla Site Controller also establishes outbound-initiated connections to Tesla Servers using reverse HTTPS proxy via mutually authenticated TLS. This is achieved using a unique X.509 certificate on the Tesla Site Controller. Finally, the Tesla Site Controller establishes outbound-initiated connections with a third-party server that hosts firmware update files. The firmware update files are encrypted and signed; the Tesla Site Controller decrypts and verifies the signature of the firmware files before using them.

Aside from the connections described in this section, the Tesla Site Controller does not establish any other connections.

Standard Data Flows on page 11 below illustrates Tesla's standard solution architecture. The *outbound-initiated connections* are made through the cellular modem or the customer-provided hardline internet connection. Connections to other customer-facing services hosted on the Tesla Site Controller (*SCADA RTU* and *UI Client*) are made over Customer Network and the LAN 1 Ethernet port as detailed in *Tesla Site Controller Logical Ports for Customer-Facing Services* on page 14.

Figure 3. Standard Data Flows



3.2.1.1 Standard Connection via the Cellular Modem

As discussed above, the Tesla Site Controller's integrated firewall only permits outbound-initiated connections on the interface with the integrated cellular modem. The only traffic that is permitted on this interface is traffic on established connections and pings (icmp echo-repy and icmp echo-request).

3.2.1.2 Standard Connection via the Tesla Site Controller's LAN 1 Ethernet Port

The Tesla Site Controller's integrated firewall permits new inbound SSH connections (for troubleshooting and local changes) on the LAN 1 Ethernet port, but only from addresses that are identified as private by the Internet Assigned Numbers Authority. SSH connections require certificates; Tesla only uses time-limited certificates for production devices and only grants certificates to a limited number of users, all of whom must be qualified Tesla employees

3.2.1.3 Through a Customer Firewall

Some customers are willing to provide the Tesla Site Controller's LAN 1 Ethernet port with a perpetual connection to the internet via a hardline connection but require a firewall on the internet connection. In this case, the customer must provide the firewall and must ensure that the ports as indicated below are open for Tesla's remote access and data logging.

The firewall must be capable of URL filtering to ensure traffic is routed correctly to and from the Tesla Site Controller.

Table 3. Firewall Rules for Tesla Remote Access and Data Logging

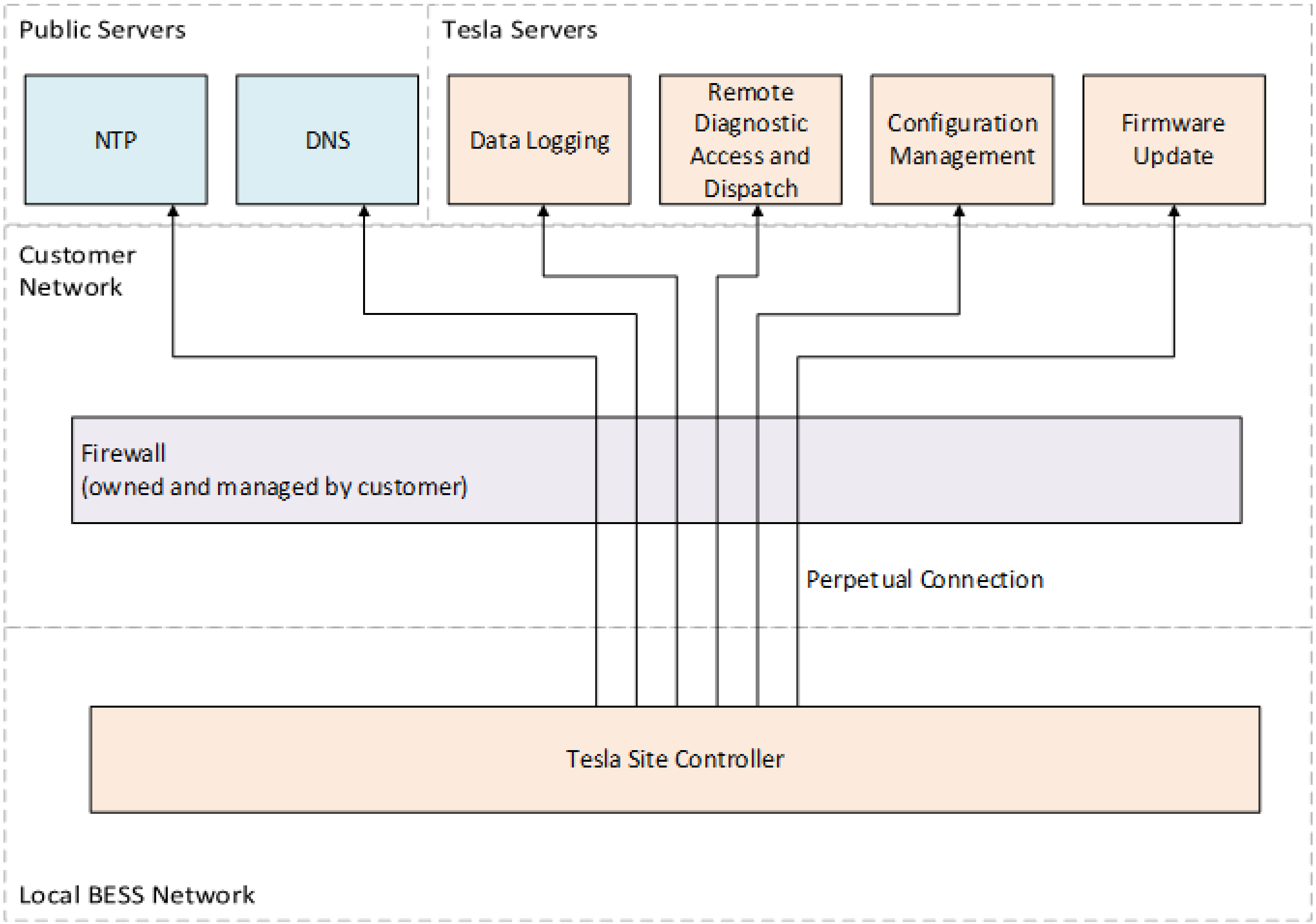
Service	Hostname	TCP/UDP	Logical Port
Logging	Two hosts are required: • https://synergy.sn.teslaenergy.services • https://hermes-stream-prd.sn.tesla.services	TCP	443
Remote Login	https://hermes-prd.sn.tesla.services	TCP	443
Configuration Files	https://configupdate-prd.sn.tesla.services	TCP	443
Firmware Update	https://maestro-prd.sn.tesla.services	TCP	443
Firmware Update	http://s3-us-west-2.amazonaws.com	TCP	80
NTP	pool.ntp.org	TCP & UDP	123
DNS	OpenDNS servers: • 9.9.9.9 • 208.67.222.222 • 208.67.220.220	TCP & UDP	53

If a customer provides a hardline internet connection through a firewall, and all of the logical ports in are perpetually open, then Tesla can disable the Tesla Site Controller cellular modem.

To enable Tesla Site Controller's data logging traffic via a proxy server through a customer firewall, refer to *Data Logging via Proxy* on page 15.

The figure below shows the remote connections that are required through the customer firewall:

Figure 4. Remote Connection Through Customer Firewall



It is also the customer's responsibility to whitelist the Tesla Site Controller logical ports for any necessary customer-facing services and applications hosted on the Tesla Site Controller if intended to be accessed through the customer firewall:

Table 4. Tesla Site Controller Logical Ports for Customer-Facing Services

Logical Port	TCP/UDP	Application
80	TCP	redirected to port 443/GUI
443	TCP	GUI
10443	TCP	REST API
502	TCP	MODBUS
20000	TCP	DNP3

3.2.2 Alternative Remote Connection Options

Customers who are unable to provide the Tesla Site Controller with an uninterrupted, perpetual connection must provide an access pathway for each of the following:

- Data Logging
- Remote Login and File Transfer
- DNS Resolution

- Time Synchronization

The following sections describe alternative options for each of these. If the customer provides an acceptable access pathway for each of these services, then Tesla can disable the Tesla Site Controller cellular modem.

3.2.2.1 Data Logging via Proxy

If the customer cannot provide the Tesla Site Controller with perpetual access to the logging server, then the customer must regularly retrieve the logs from the Tesla Site Controller and upload them to the logging server at least as often as once every minute. Once uploaded, Tesla's data infrastructure must reprocess the data, which typically can take up to two minutes. This delay will impact the data's availability to systems, applications, and user interfaces that rely on the Tesla servers for this data, such as Powerhub.

Tesla can provide a proxy application that the customer can use for periodic log retrieval and upload. The proxy application must be installed on the Customer Network. The proxy application requests encrypted logs from the local Tesla Site Controller, receives the signed and encrypted log files, and then pushes the data logs to the Tesla server. This must be an uninterrupted perpetual connection.

Tesla can provide the source code of the proxy application to the customer for review, if needed. Tesla can provide an executable file compiled from the source code, if desired.

The customer is responsible for installing and maintaining the network infrastructure and computing environment that hosts the proxy application, also known as the “proxy server.” The availability of the infrastructure will also be the customer’s responsibility. The proxy server must meet the following minimum requirements:

- RAM: 8 GB
- Processor: Single core 2 GHz
- Hard Drive Space Available: 128 GB
- Available 100/1000MB (autonegotiating) Ethernet Port
- Up-to-date Windows operating system
- Ongoing support to apply required security patches to operating system

The customer must configure the firewall between the Tesla Site Controller and the proxy server (if present) to allow the connection listed in the table below from the proxy server to the Tesla Site Controller. The Tesla Site Controller will authenticate incoming log requests from the proxy application and establish a secure connection:

Table 5. Firewall Rules Between Proxy Server and Tesla Site Controller

Service	Hostname	TCP/UDP	Logical Port
Logging	Tesla Site Controller LAN1 IP address	TCP	443

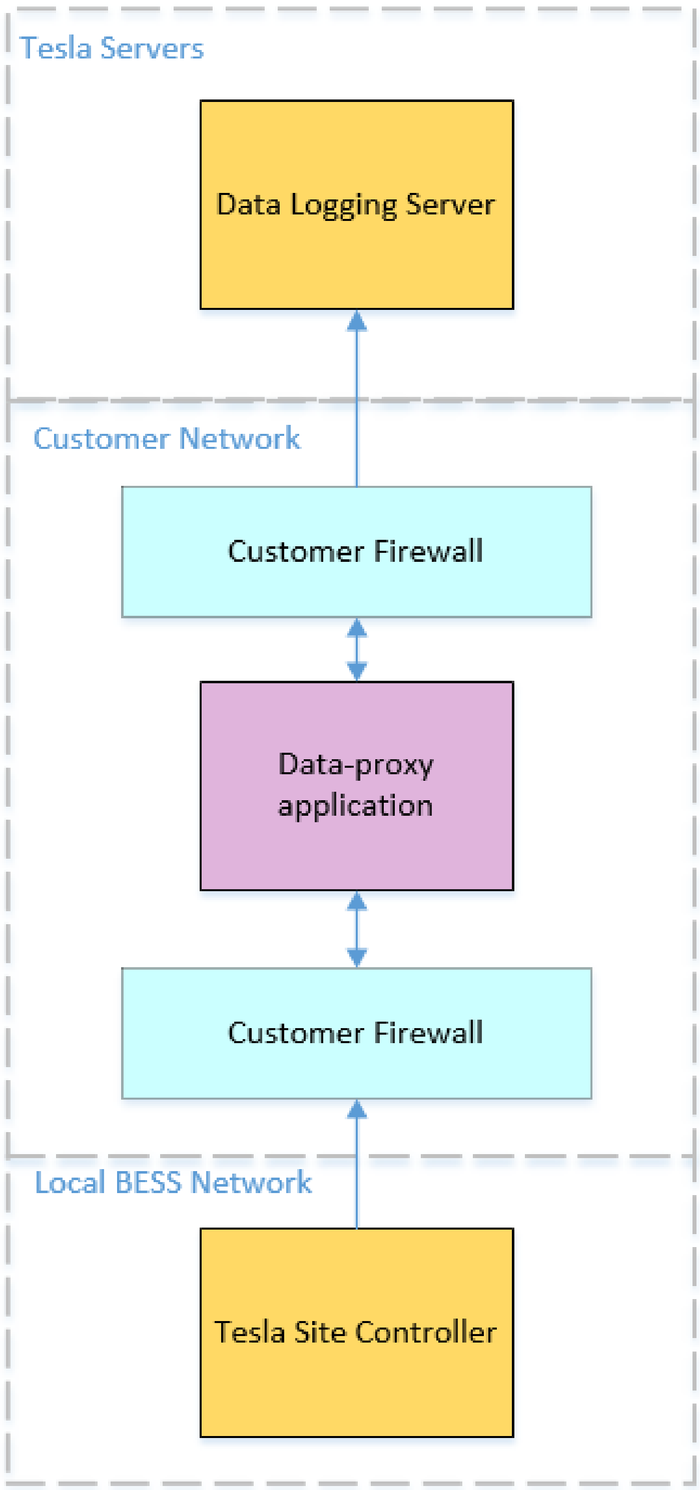
The customer must also configure the firewall between the proxy server and the internet to allow the outbound connections listed in the below. The proxy application and the server use mutual TLS and encryption:

Table 6. Firewall Rules Between Proxy Server and Internet

Service	Hostname	TCP/UDP	Logical Port
Logging	*https://log-uploader.sn.tesla.services*	TCP	443

See the figure below for more details:

Figure 5. Firewall Rules with Proxy Server



For more information on commissioning the data proxy application, see *Appendix B: Data Proxy Commissioning* on page 35.

3.2.2.2 Remote Login and File Transfer

Remote login is required for diagnostics and remote dispatch, while remote file transfer is required for configuration management and firmware updates. If the customer cannot provide the Tesla Site Controller with perpetual access to the servers used for remote diagnostic access and dispatch, configuration management, and firmware update, then the customer must provide one of the following solutions:

- Access On-Request
- Access via Jump Server

The two options are outlined below:

Access On-Request

When the Tesla Site Controller uses a customer-provided hardline internet for remote login and file transfer, the customer can choose to make this connection unavailable in regular operation and only make it available on request by Tesla. In this case, Tesla-authorized personnel will contact the customer for each remote login and file transfer session. Any delay or wait time in making the connection available will not count toward Tesla's "downtime for availability" or any other system downtime metric calculated in any warranties or performance guarantees between the customer and Tesla. Furthermore, any damage to the system due to delayed access will be the liability of the customer and not of Tesla.

Once the customer makes the connection available for remote login and file transfer, the customer firewall must allow outbound connections from the Tesla Site Controller to the internet in order to initiate a connection to the services listed in *Firewall Rules for On-Request Access* on page 17. The connection from the Tesla Site Controller to Tesla servers is made using reverse HTTPS proxy via mutually authenticated TLS. This is achieved using a unique X.509 certificate on the Tesla Site Controller, and this connection is only used to transmit signed and encrypted firmware update files. The Tesla Site Controller decrypts and verifies the signature of the firmware files before using them.

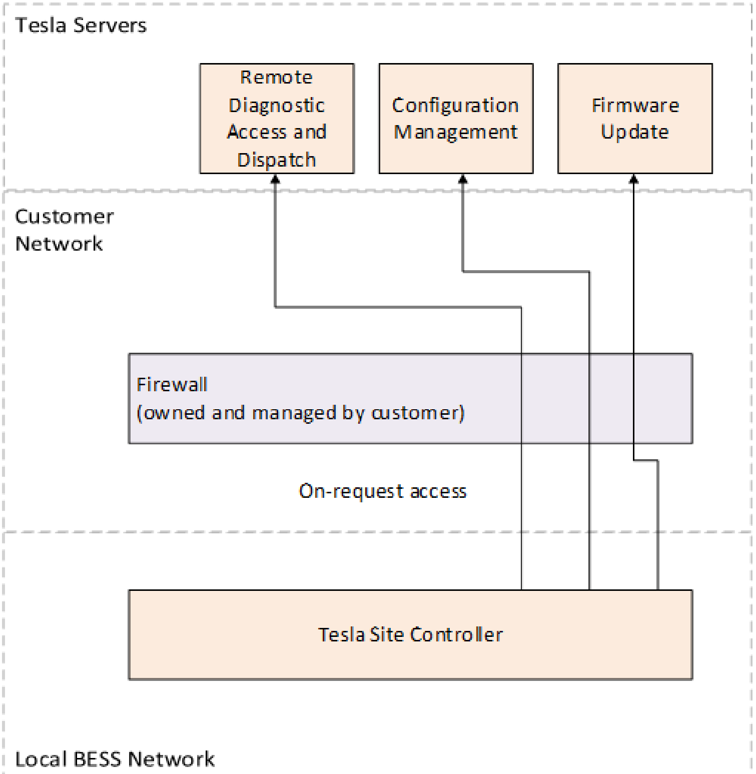
Table 7. Firewall Rules for On-Request Access

Service	Hostname	TCP/UDP	Logical Port
Remote Diagnostic Access and Dispatch	*https://hermes-prd.sn.tesla.services*	TCP	443
Configuration Management	*https://configupdate-prd.sn.tesla.services*	TCP	443
Firmware Update	*https://maestro-prd.sn.tesla.services*	TCP	443
Firmware Update	*http://s3-us-west-2.amazonaws.com*	TCP	80

In order to resolve hostnames for remote login, a DNS server must be available at the time that on-request access is granted. This is explained in greater detail in *DNS Resolution* on page 20.

The on-request access alternative is shown in the following figure:

Figure 6. On-Request Access Network



Access via Jump Server

If the customer is unable to open the firewall logical ports listed in *Firewall Rules for On-Request Access* on page 17, then the customer must provide Tesla with VPN access to a customer-provided jump server on the same local network as the Tesla Site Controller. Tesla will use the jump server for remote diagnostic access and dispatch, configuration management, and firmware updates.

Tesla's approximate procedure for securely accessing the Tesla Site Controller via a jump server is as follows. Tesla will generate and sign a certificate, transfer it to the jump server, and use that certificate to SSH to the Tesla Site Controller.

The jump server must meet the following minimum requirements:

- RAM: 8 GB
- Processor: Single core 2 GHz
- Hard drive space available: 128 GB
- Available 100/1000 MB (autonegotiating) Ethernet Port
- Up-to-date Windows operating system
- Ongoing support to apply required security patches to operating system
- OpenSSH
- Google Chrome browser
- File transfer capability
- Other third-party apps like RTAC AcSELerator, relay, meter apps (as specified by Tesla on a per-project basis)
- Tesla requires admin privileges

The customer must provide and maintain the VPN service and provide Tesla with VPN access credentials. The VPN service must permit file transfers between Tesla and the jump server. Any VPN downtime will not count toward Tesla's "downtime for availability" or any other system downtime metric calculated in any warranties or performance guarantees between the customer and Tesla. Furthermore, any damage to the system due to VPN downtime will be the liability of the customer and not the liability of Tesla.

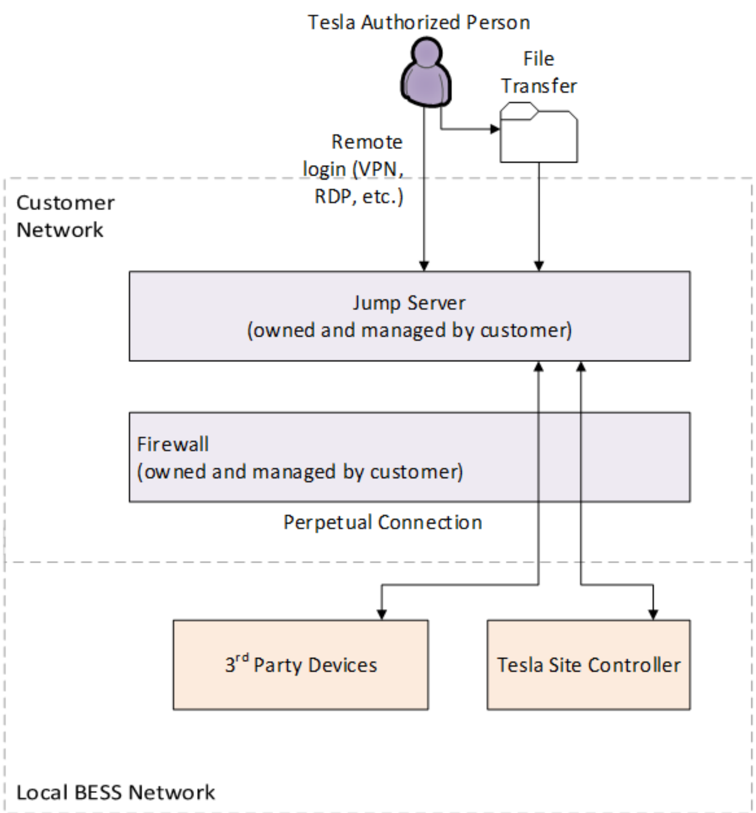
If there is a firewall between the jump server and the Tesla Site Controller, the firewall must have the following logical ports open:

Table 8. Firewall Rules Between Jump Server and Tesla Site Controller

Service	Hostname	TCP/UDP	Logical Port
SSH, SCP	Jump server IP address	TCP	22
HTTPS	Jump server IP address	TCP	443

If the project has other third-party devices on LAN1, such as an SEL RTAC, additional logical ports will need to be open on the firewall.

Figure 7. Jump Server Network



3.2.2.3 Time Synchronization

The Tesla Site Controller requires an external time source in order to correctly timestamp logged data. By default, the Tesla Site Controller uses a public NTP server for time synchronization.

If the customer cannot provide the Tesla Site Controller with perpetual access to a public NTP server, then the customer must provide an NTP server on the local network. This will be typical for sites where the customer provides remote access via a jump server. For hardware guidance, see *Time Synchronization on page 32*.

3.2.2.4 DNS Resolution

DNS resolution is used for log upload, remote login, and time synchronization.

The Tesla Site Controller requires access to a DNS server under the following conditions:

- If the Tesla Site Controller is logging directly through the customer firewall (not via proxy)
- At the time that on-request access is granted, if remote login and file transfer is provided via on-request access
- If time synchronization is using a public NTP server through the customer firewall (not via a local time source)

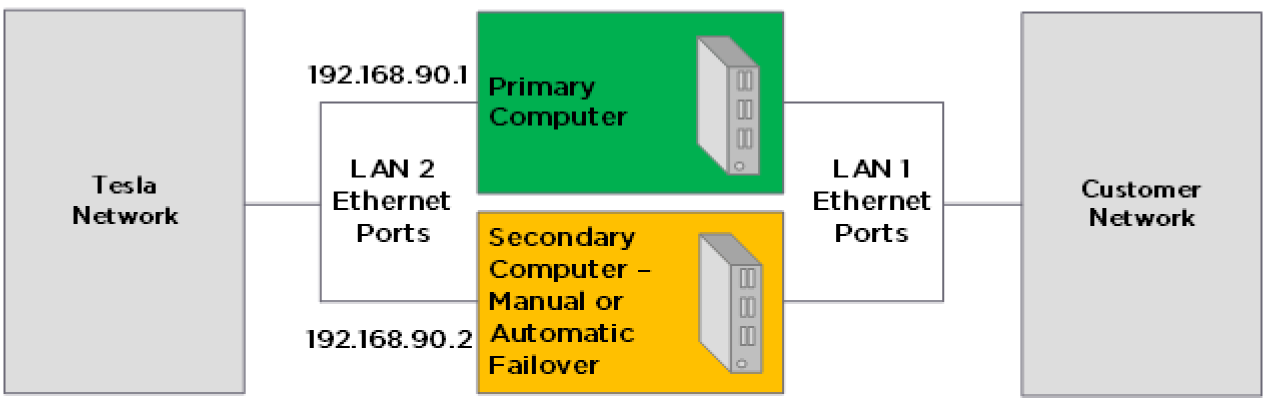
If logging is being performed via proxy, the proxy server needs access to a DNS server.

If the customer cannot provide the Tesla Site Controller and proxy server with perpetual access to a public DNS server, the customer must provide a local DNS server that is regularly synced with a public DNS server.

3.3 Network Redundancy

To improve availability of the battery system and to prevent downtime, Tesla has developed redundant network architecture standards for the Tesla Site Controller.

Figure 8. Tesla Site Controller Redundancy: Manual or Automatic



Tesla, by default, includes a spare Tesla Site Controller that can be utilized for redundancy purposes when installed and configured as detailed in the figure above. Check with your Tesla team if redundancy is required for your project. Tesla's offerings for Manual Failover and Automatic Failover are described below.

Manual Failover: A secondary Tesla Site Controller is installed and connected to both power and communications. In the event of a failure of the primary Tesla Site Controller, Tesla personnel will manually and remotely login and re-establish site operation with the secondary Tesla Site Controller.

Automatic Failover: A secondary Tesla Site Controller is installed and connected to both power and communications. During normal operation, the secondary Tesla Site Controller is configured to remain inactive unless there is a failure of the primary Tesla Site Controller. In the event of a primary controller failure, site control is seamlessly transferred to the secondary controller without manual intervention.

4 Selecting SCADA Hardware

This section describes Tesla's requirements and recommendations for the selection and application of SCADA equipment.

The major SCADA hardware/network components involved in a typical project may include but are not limited to:

- *Battery Inverters* on page 21
- *Tesla Site Controller* on page 21
- *Meters* on page 23
- *SEL RTAC* on page 23
- *RTAC Applications* on page 24
- *Network Switches* on page 25
- *SCADA Enclosures* on page 26
- *Fiber Optic* on page 31
- *Fiber-to-Ethernet Conversion* on page 32
- *Serial-to-Ethernet Conversion* on page 32
- *Network Firewall* on page 32
- *I/O (Input/Output) Devices* on page 32
- *Time Synchronization* on page 32
- *HMI/GUI Display* on page 33

Each component is described below.

4.1 Battery Inverters

Tesla battery inverters are required to be connected to the Tesla Site Controller over the Tesla Network. The Tesla Site Controller functions as the DHCP server that assigns IP addresses to the battery inverters and uses a proprietary TCP protocol for communication.

4.2 Tesla Site Controller

The Tesla Site Controller is the single point of interface for the utility, network operator, or customer SCADA systems to control and monitor the entire energy storage site. It hosts the control algorithm that manages the charge and discharge functions of the battery system units, aggregating real-time information and using the information to optimize the commands sent to each individual battery unit.

The battery system communicates with the Tesla Site Controller over a private TCP network. Customer systems communicate with the Tesla Site Controller via Modbus, DNP3, or REST API interfaces.

One Tesla Site Controller is typically required for each point of interconnection, however Tesla may choose to provide a second redundant Tesla Site Controller. Tesla does not include additional networking equipment that may be required for the system to operate.

The Tesla Site Controller has three network interfaces:

- LAN 2 RJ45/Ethernet port, which connects to the Tesla Network

- LAN 1 RJ45/Ethernet port, which connects to the Customer Network and can be configured for WAN access
- Integrated cellular modem, which provides cellular access for Tesla's remote connection

There are two physical variations of the Tesla Site Controller:

- **Standard Tesla Site Controller -** used in sites up to 18 MW. Delivered in an enclosure called the Standard Tesla Site Controller Enclosure, which may include one or two Standard Tesla Site Controllers.
    - DIN-rail-mounted, embedded Linux industrial PC: *http://www.axiomtek.com/Download/Spec/en-US/ico300.pdf*

- **Large Tesla Site Controller -** for sites larger than 18 MW. Delivered as the controller only.
    - Rack- or shelf-mounted PC: *http://www.cincoze.com/data/files/201809/Datasheet_DS-1100_R20.pdf*

Figure 9. Standard Tesla Site Controller in the Standard Tesla Site Controller Enclosure

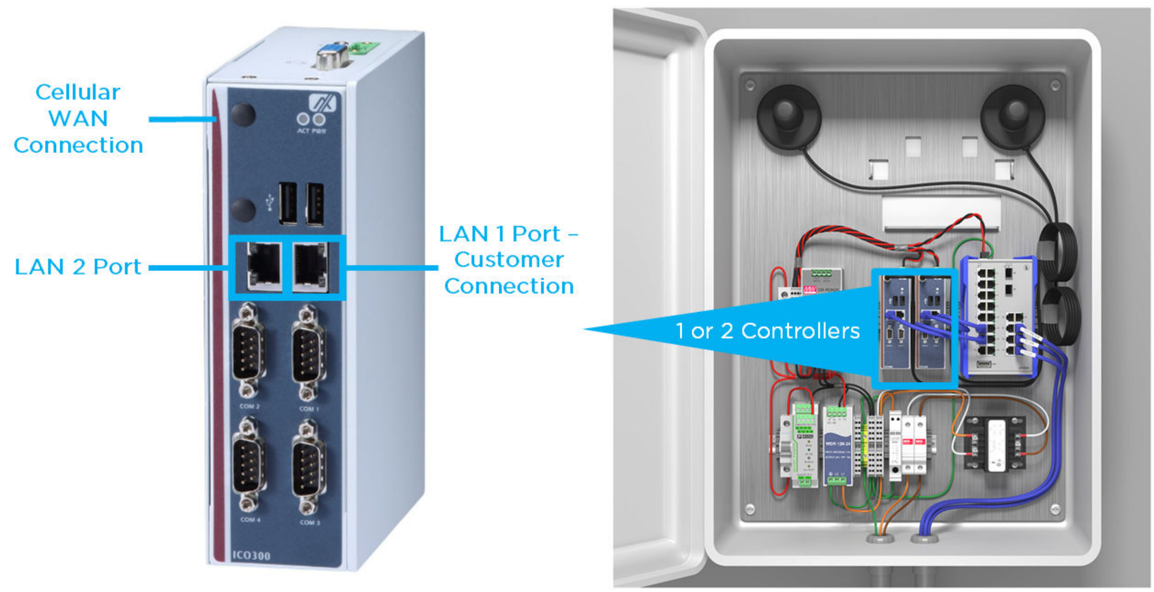
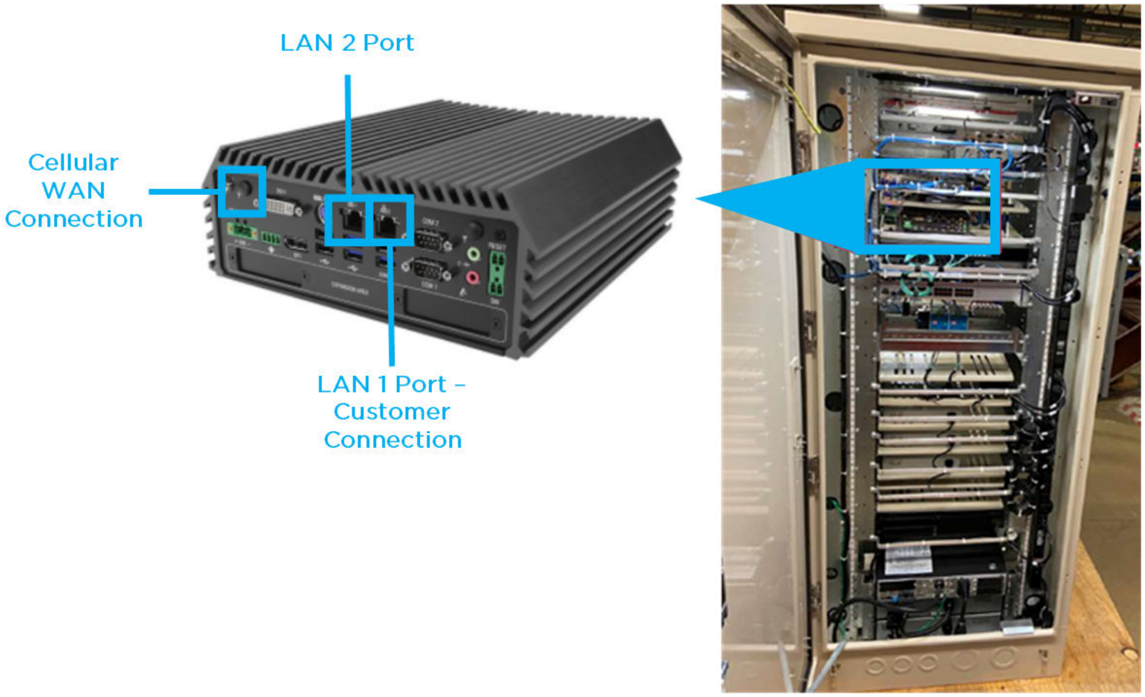


Figure 10. Large Tesla Site Controller mounted inside SCADA Rack Enclosure



> **NOTE:** DB9 and USB ports are disabled and not used for any communication on the devices.

For more information:

- See the *Tesla Energy Controls and Communications Manual* for complete instructions on how to interface with the Tesla Site Controller.

4.3 Meters

The Tesla Site Controller communicates directly with and relies on various meter inputs for the different control functions described in the *Tesla Energy Controls and Communications Manual*. Refer to the appropriate *Site Design Manual* for details on Tesla's meter selection requirements.

4.4 SEL RTAC

This section outlines Tesla's internal guidance for using an RTAC to meet contractual requirements. A Tesla-managed RTAC is *not* a standard offering, and additional costs are associated with RTAC hardware, licensing, and engineering.

4.4.1 Hardware Selection

RTAC is the Real-Time Automation Controller by Schweitzer and is widely used on utility and commercial/industrial sites. RTAC is available in a few different hardware variations and form factors such as RTAC 3530 and 3555. Tesla typically uses the RTAC 3530-4, which has two LAN ports and four serial ports. For projects that need more physical communication ports or have more than 25,000 tags, Tesla typically uses the 3555.

Figure 11. SEL RTAC 3530-4 (Front and Back)

4.4.2 RTAC Connection and Interfacing

The Tesla-managed RTAC may be connected to both the Tesla Network and the Customer Network.

Customers can interface with the Tesla-managed RTAC over the Customer Network and may use the Modbus or DNP3 communication protocol over TCP. Customers can also connect to the RTAC using a Serial RS485/RS232 connection to access data using the Modbus or DNP3 communication protocol.

4.4.3 RTAC Applications

The following are the most common applications for which Tesla may use an RTAC to meet contractual obligations.

Custom Points List

When a customer requires a custom points list that is different from what is available directly via the TSC's Modbus and DNP3 interfaces, the RTAC can be programmed to read from the TSC and relay that data to the customer in a custom-formatted points list. The RTAC does not support the REST API and therefore cannot be used to create a custom REST API interface.

For example, the RTAC may perform:

- Custom scaling
- Custom addressing of points
- Custom data format (example holding register vs input register)
- Custom communication protocols (example IEC 61850)

Non-Standard Equipment Telemetry

Tesla may use the RTAC as a data concentrator to aggregate data from non-standard equipment such as circuit breaker status, relay data, or fire alarm status. If the equipment is located on the Tesla Network, a Tesla-managed RTAC can be used to gather data from the Tesla Network and pass it to the customer over the Customer Network.

Control Arbitration

The Tesla Site Controller is not capable of assigning priority to multiple command sources and can only react to the most recent command. When more than one control interface is active (for example: customer RTU, Powerhub Local, ISO RTU), an RTAC can be configured as an arbitrator and can restrict access to the TSC from external controllers (for example: customer RTU, ISO RTU) based on a custom logic scheme.

> **NOTE:** The TSC can be configured to send a flag to an external Modbus server's holding registers to communicate when the Powerhub Local interface is set to remote/local. This can be used to allow customer arbitration between their controller and Powerhub Local control.

Custom Controls

For any the custom control functionality outside of standard control modes listed in the *Tesla Energy Controls and Communication Manual*, the RTAC may be used to implement the required control logic. Any custom control functionality requirements must be agreed to and documented during the contracting phase of a project. Some of the common examples of control modes/functions that can be programmed in the RTAC are:

- Trip and close of a circuit breaker
- Start and stop of ancillary loads/generators
- Simple logic to relay certain data for certain pre-specified conditions

4.5 Network Switches

Tesla *recommends* the use of these switches and requires them for the specified use cases described in this section for which Tesla has a contractual obligation for implementation, operation, and maintenance of the network.

The Cisco IE5000 and Cisco IE3400 are used as layer 2 switches that facilitate:

- Configuration management and network traffic monitoring
- Support of VLANs, syslog, and network ring architecture
- Accommodation of fiber terminations, when specified with the appropriate SFP ports and transceivers

Cisco IE3400 may be used in the field network enclosure (FNE). Cisco IE5000 may be used as the core switch for the Tesla Network and for the Customer Network inside the Point of Interconnection (POI) enclosure.

> **NOTE:** Tesla may add or remove part numbers as new switch product releases become available. Please check with your Tesla Project Engineer or Tesla SCADA Engineer to confirm that the required switches are the latest guidance at the time of your design, or if any other switch type for the project is required.

4.5.1 Cisco IE3400 Switch

The Cisco IE3400 switch is typically used in each FNE as shown in *Network Architecture* on page 8.

Cisco IE3400 is a rugged, heavy-duty, DIN-rail mountable switch. The base configuration (IE-3400-8T2S-A) comes with 2 x 100/1000 SFP-based ports and 8 x 10/100/1000 RJ45 copper ports.

Figure 12. Example IE3400 Switch

4.5.2 Cisco IE5000 Switch

The Cisco IE5000 switch is typically utilized as the core switch as shown in *Network Architecture* on page 8. Typically, it is located in the centralized SCADA rack/enclosure and is directly connected to the LAN 2 ethernet port of the Tesla Site Controller. The IE5000 is also recommended for use as the core Customer Network switch.

Cisco IE5000 is a rack-mounted and ruggedized switch. The base configuration (IE5000-12S12P-10G) comes with 12 x 1G SFP-based ports and 12 x 10/100/1000 + 4 x 1G/10G RJ45 copper ports.

Note that per Cisco's installation guidelines, to prevent overheating the IE5000 switch must have a minimum clearance of 4.4 cm (1.75 in) on the top and bottom.

Figure 13. Example IE5000 Switch

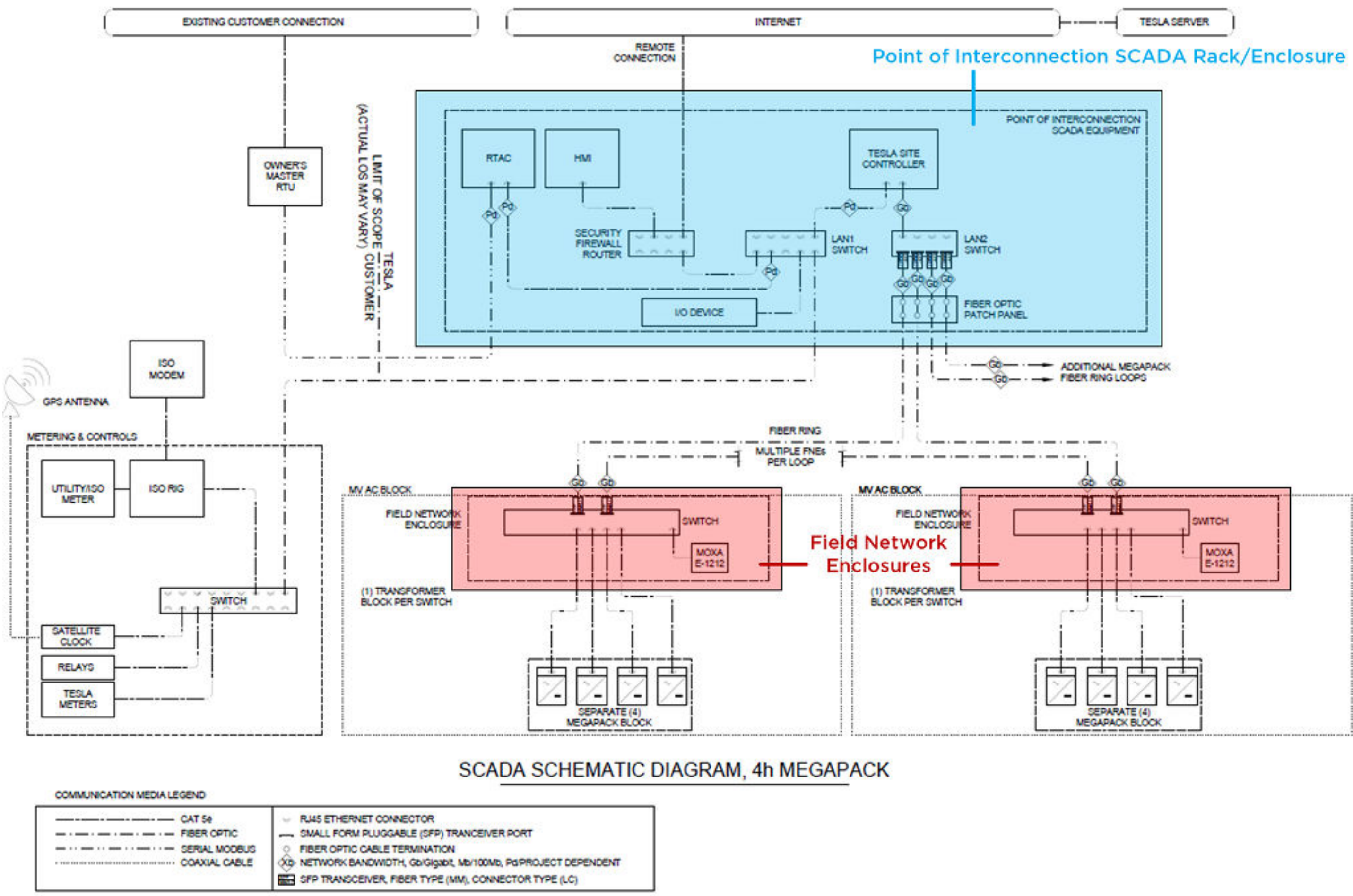
4.6 SCADA Enclosures

This section outlines Tesla’s internal guidance for the standardization of SCADA Enclosures. Design and supply of these enclosure is non-standard, contact your Tesla representative for more information..

For large-scale utility projects, Tesla has standardized the network architecture around two networking nodes.

The figure below shows a standard SCADA architecture diagram and identifies these two network nodes for a typical site.

Figure 14. Standard SCADA Enclosures — Example Layout



4.6.1 POI SCADA Rack / Enclosure

This SCADA rack typically houses the Tesla Site Controller and core network switches. It may also house the RTAC, firewall, I/O unit(s), UPS, GPS clock, HMI, Historian, and/or any other co-located SCADA equipment. The SCADA rack can be designed in an enclosure for security, environmental protection, and for other project specific conditions or requirements.

Figure 15. Example Indoor SCADA Rack

Figure 16. Example Outdoor Temperature-Controlled SCADA Rack Enclosure - Open

Figure 17. Example Outdoor Temperature-Controlled SCADA Rack Enclosure - Closed

4.6.2 Field Network Enclosure

The Field Network Enclosure (FNE) is an outdoor-rated (NEMA/UL type 4) enclosure designed to host the networking equipment required for each medium-voltage (MV) AC block including the network switch, an I/O unit, fiber optic patch panel, and power electronics. The network switch on the FNE is part of the site's communication network backbone. The FNE is a modular approach to SCADA system design.

Field Network Enclosures should not be placed in direct sunlight whenever possible. It is recommended that enclosures are located on the north side (Northern Hemisphere) of equipment or should be provided with shade structures on any site where direct sunlight cannot be avoided. The engineer/designer is responsible to ensure that the enclosure's internal temperature does not exceed the enclosed device's ratings.

As shown in the *SCADA Architecture Diagrams* and as is typical for large sites, BESS inverters in a MV AC block are connected radially to the network switch inside the FNE. The network switches within the FNE are connected in a fiber ring architecture. The fiber ring architecture is utilized for improved reliability and to avoid single-point failure scenarios. All rings should terminate at the core switch in the POI SCADA Enclosure.

Alternative architectures should be approved by a Tesla SCADA Engineer on a case-by-case basis.

Figure 18. Fiber Ring Architecture

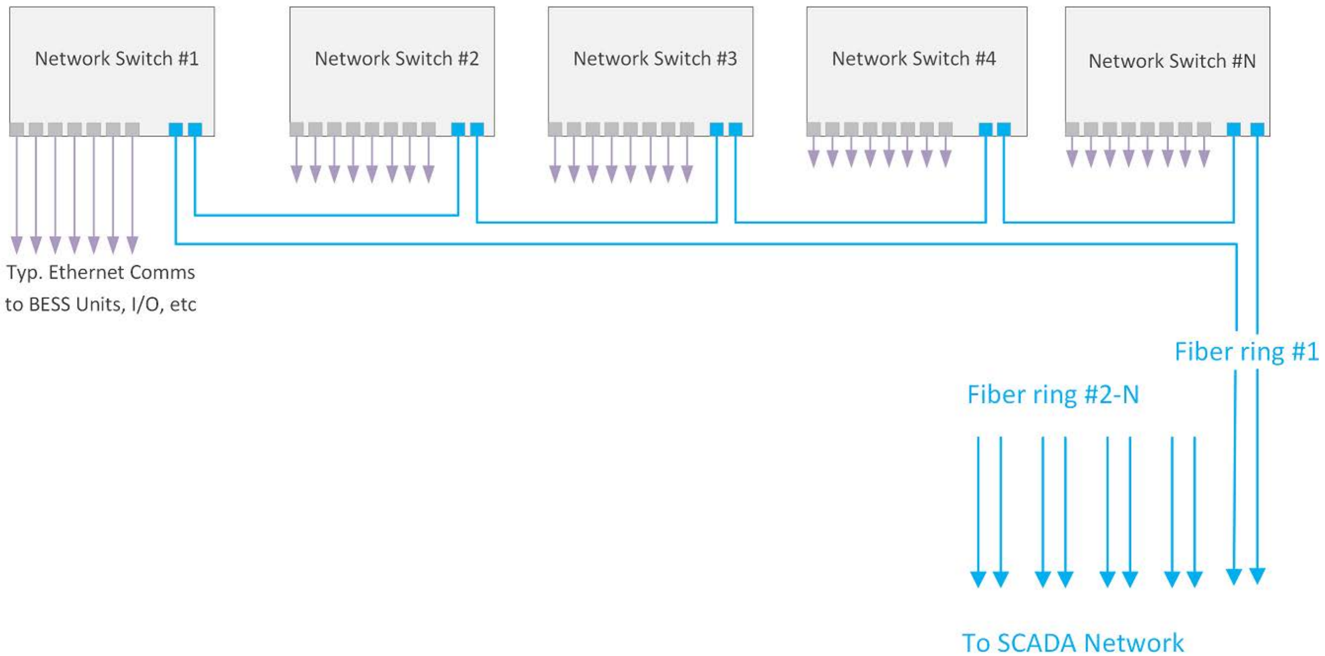
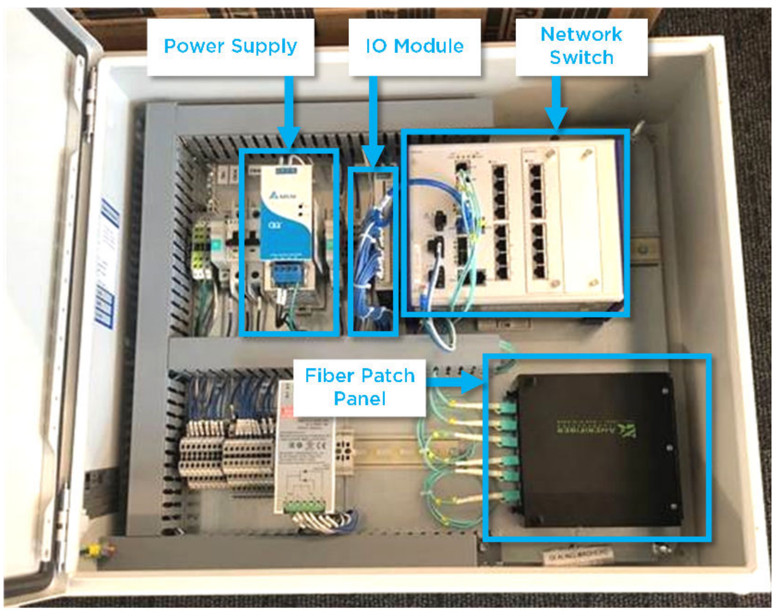


Figure 19. Field Network Enclosure Components



4.7 Fiber Optic

Tesla recommends the use of fiber optic communications when cable distance exceeds 90 meters or if higher network reliability is required due to the size and criticality of the site.

Tesla does not specify the make/model for the fiber cable. However, Tesla recommends the use of single-mode duplex fiber cable types and LC terminations. When fiber is pulled through conduit, patch panels or splice closures should be used at each end, and the cables must be tested by a qualified fiber optic contractor before system commissioning.

4.8 Fiber-to-Ethernet Conversion

Where fiber-to-Ethernet conversion is necessary, Tesla recommends the Hirschmann-SPIDER-1TX/1FX media converter. Note that this device uses SC fiber terminations as opposed to the Tesla-standard LC terminations.

4.9 Serial-to-Ethernet Conversion

Where serial-to-Ethernet conversion is necessary, Tesla recommends the Moxa MB3170 and MB3270 Ethernet-to-serial adapters. These devices do not support DNP3 communication protocol.

4.10 Network Firewall

When firewall operation and management is within Tesla's scope, the Juniper SRX-3xx series is required to be used; outside of Tesla's scope it is recommended.

4.11 I/O (Input/Output) Devices

Tesla offers integration of some I/O devices directly with the Tesla Site Controller. For more information, refer to the document *Configuration Specification: Using I/O Devices with the Tesla Site Controller*.

Figure 20. Example I/O Devices: Emerson (Left) and Moxa ioLogik E1212 (Right)

4.12 Time Synchronization

As described in *Tesla Remote Connection* on page 10, the Tesla Site Controller requires access to an NTP server for time synchronization. Connected devices can be configured to use the Tesla Site Controller as an NTP server, for example meters, network switches, etc.

4.12.1 SEL-2488 Network Clock

Where a separate network clock is necessary, Tesla recommends using the SEL-2488 with the SEL-9524 antenna accessory.

The SEL-2488 receives Global Navigation Satellite System (GNSS) time signals and distributes precise time signals via multiple output protocols, including IRIG-B and the Network Time Protocol (NTP). NTP provides ~100µs accuracy with easy setup time and the ability to use an existing Ethernet network communication backbone without separate wiring. SEL-2488 is configured with an NTP server on each Ethernet port. To synchronize the device time over Ethernet, each device subscribes to the NTP server's time signals over a set IP address. For higher accuracy requirements up to ~100ns, IRIG-B time synchronization output connections are also supported by SEL-2488 hardware, which requires separate coaxial twisted-pair cabling.

Figure 21. Example SEL-2488 Time Synchronization Clock

4.13 HMI/GUI Display

A computer display is often required to access or host HMI/GUI software. See the *Powerhub User Manual* for instructions on how to access the Powerhub GUI via a computer and web browser.

Appendix A: Standard IP Address Range

Table 9. Tesla Network IP Address Range

Device	DHCP Status	IP Range Start (IP Address If Static)	IP Range End	Subnet	Comment
Battery Meters	Static	192.168.90.201	192.168.90.209	255.255.240.0	Provision for 10 meters
Busway Meters	Static	192.168.90.230	192.168.90.239	255.255.240.0	Provisions for 10 meters
Generator Meters	Static	192.168.90.240	192.168.90.249	255.255.240.0	Provisions for 10 meters
Tesla Site Controller	Server	192.168.90.1	192.168.90.3	255.255.240.0	-
PV Meters	Static	192.168.90.220	192.168.90.229	255.255.240.0	Provisions for 10 meters
RTAC	Static	192.168.90.04	192.168.90.05	255.255.240.0	
SEL700G	Static	192.168.90.50	-	255.255.240.0	
Site Meter	Static	192.168.90.210	192.168.90.219	255.255.240.0	Provisionsfor 10 meters
Tesla Inverters	Client	192.168.80.1	192.168.89.254	255.255.240.0	Provisions for2640
Tesla Inverters	Client	192.168.92.1	192.168.95.254	255.255.240.0	Provisions for1016
FNE switches	Static	192.168.91.1	192.168.91.100	255.255.240.0	Provisions for100 FNEs
I/O Device	Static	192.168.91.101	192.168.91.200	255.255.240.0	Provisions for100 IOs
Other Switches	Static	192.168.91.51	192.168.91.70	255.255.240.0	Provisions for 20 switches

Appendix B: Data Proxy Commissioning

General Recommendations

- To avoid interruptions to data upload, ensure that the data proxy server runs continuously and is not interrupted by things such as hardware reboots. Commonly, the data proxy is run as a “windows service” to ensure automatic restart of the data proxy application.

Deploying and Running the Data Proxy Application

- Get a data proxy application and the associated username and password by reaching out to your Tesla engineering team.
- Move the data proxy application to the proxy server.
- Run the data proxy application by executing the file with the proper modifying parameters. Typically:
    - -host string
        - Base URL of the host server where logs are stored (default "https://localhost:443")
    - - loop duration
        - Will upload logs at every provided interval
        - For example “7m” = every seven minutes, “4h” = every four hours
    - -state-dir string
        - data proxy state directory (default "/var/lib/data-proxy")
        - directory where logs are saved prior to upload
    - -username string
        - Username to use to authenticate (certificate takes precedence)
        - Username is provided by Tesla
    - -password string
        - Password to use to authenticate
        - Password is provided by Tesla
- Example run command:
    - ./data-proxy.exe -host "https://192.168.1.1:443" -loop 1m -state-dir "C:\proxy-logs" -username <username> -password <password>
    - Where <username> is replaced with the Tesla provided username and <password> is replaced with the Tesla provided password
    - Where 192.168.1.1 is the IP address of the Tesla Site Controller’s LAN 1 ethernet port
    - Where the logs will be saved to the directory “C:\proxy-logs”
    - Where you want the logs uploaded every 1 minute (default required by Tesla)
    - Where the logs will be saved to “C:\proxy-logs” prior to upload
- Once the application is running it should display an operational application log file that looks similar to the below snapshot when running correctly. The operational application log is a useful troubleshooting tool that is referenced in the troubleshooting section:

```
2021/07/15 23:31:57.233893 Considering 15 files
2021/07/15 23:31:57.233893 Processing 15 files
2021/07/15 23:31:59.392817 Uploaded STST-SM-    _2021-06-30T06_04_03_TG1203570024S0_STARC_MP_DRLog.blf in 1.9859777s
2021/07/15 23:32:01.388359 Uploaded STST-SM-    _2021-06-30T10_32_38_TG120357000CS4_STARC_a022_FPGA_HV_OV.blf in 1.8478815s
2021/07/15 23:32:03.467478 Uploaded STST-SM-    _2021-07-03T21_15_44_TG120357000CS4_STARC_a022_FPGA_HV_OV.blf in 1.9446231s
2021/07/15 23:32:05.666623 Uploaded STST-SM-    _2021-07-05T09_00_25_TG120357000CS4_STITCH_DRLog.blf in 2.0554816s
2021/07/15 23:32:07.943125 Uploaded STST-SM-    _2021-07-06T01_20_40_TG120357000CS4_STARC_a022_FPGA_HV_OV.blf in 2.1484661s
2021/07/15 23:32:09.971339 Uploaded STST-SM-    _2021-07-09T06_35_39_TG120356001Z16_STARC_a022_FPGA_HV_OV.blf in 1.8818737s
2021/07/15 23:32:12.071020 Uploaded STST-SM-    _2021-07-09T17_49_44_TG120356001Z16_STARC_a022_FPGA_HV_OV.blf in 1.9465535s
2021/07/15 23:32:14.137976 Uploaded STST-SM-    _2021-07-12T14_15_23_TG120356001Z16_STARC_MP_DRLog.blf in 1.9099483s
2021/07/15 23:32:16.757849 Uploaded STST-SM-    _2021-07-14T11_13_51_TG1203570024RY_STBC_a091_Self_Test_Failed.blf in 2.4449652s
2021/07/15 23:32:19.474832 Uploaded STST-SM-    _2021-07-14T11_13_55_TG120356001Z16_STBC_a091_Self_Test_Failed.blf in 2.5485906s
2021/07/15 23:32:22.189108 Uploaded STST-SM-    _2021-07-14T11_21_20_TG120357000CS4_INV_DRLog.blf in 2.5325981s
2021/07/15 23:32:24.938877 Uploaded STST-SM-    _2021-07-14T15_18_39_TG120357000CS4_INV_DRLog.blf in 2.5752716s
2021/07/15 23:32:27.869469 Uploaded STST-SM-    _2021-07-14T15_33_33_TG1203570024RY_INV_DRLog.blf in 2.7692213s
2021/07/15 23:32:30.498562 Uploaded STST-SM-    _2021-07-14T16_00_14_TG120356001Z16_INV_DRLog.blf in 2.4854348s
2021/07/15 23:32:32.497979 Uploaded STST-SM-    _2021-07-15T07_32_30_TG120356001Z16_STARC_MP_DRLog.blf in 1.8641765s
2021/07/15 23:32:32.499978 Successfully uploaded all logs
```

Testing and Verifying the Data Proxy

- (Tesla) Verify that the Tesla Site Controller is running and communicating to equipment onsite such as meters and BESS units.
- (EPC/ Customer) Run the proxy application and let it run for at least 20-30 minutes. Review and/or send a screenshot of the output log to Tesla to confirm the application is running properly.
- (Tesla) Verify that the data from the Tesla Site Controller is reaching the Tesla's data storage servers.

Troubleshooting Common Issues

- For help on the meaning/ format of the data proxy parameters:
  - From a terminal/ command prompt, navigate to the folder with the data proxy application
  - Execute the command ".\data-proxy.exe -h"
  - It should output a list of the modifiable parameters for use in the application execution command
- The "-upload-url string" parameter defaults to *https://log-uploader.sn.tesla.services/upload*
  - When deploying the data proxy, if it reports that it cannot reach the server at a different URL then the default, then include the following parameter in the data proxy deployment "-upload-url https://log-uploader.sn.tesla.services/upload"
- The following line of the operational application log indicates that the proxy application has failed to reach the Tesla servers:
  `2019/07/22 15:37:24.596319 Try 1/3: failed to upload "20190722_114848_1563821328_3031.gzip": failed to execute download request`
  - Verify that the server running the proxy application has access to the internet and Tesla Server. Try pasting the upload URL (https://log-uploader.sn.tesla.services/upload) in the browser.
  - If the customer is providing internet to the proxy server via a firewall, ensure that they are permitting outbound TCP access on logical port 443 to the upload URL (*https://log-uploader.sn.tesla.services/upload*)
  - Verify that the customer is providing a DNS server for hostname resolution or pointing the proxy server to a public DNS server.
- The following line of the operational application log indicates that the proxy application cannot reach the Tesla Site Controller:
  `"failed to retrieve log files list: Get https://192.168.90.1:443/api/v2/sitemaster/logs/files: context canceled"`

- Verify that you can ping the Tesla Site Controller from the server that is hosting the data proxy application.
- If there is a firewall between the data proxy server and the Tesla Site Controller, ensure that it has logical port 443 open.

Appendix C: Network Configuration of the Tesla Site Controller's LAN 1 Ethernet Port

Configure your laptop's IP address:

1. Connect to the Tesla Site Controller's LAN 2 ethernet port.
2. Configure your laptop's IP address:
   On Windows:
   - Go to **Control Panel** → **Network and Sharing Center** → **Change Adapter Settings** → **Ethernet** → **Properties** → **Internet Protocol Version 4**
   - Click **Use the following IP address** and ensure that your laptop's IP address is set to 192.168.90.251 and that the subnet mask is set correctly. The typical subnet mask is 255.255.255.0.

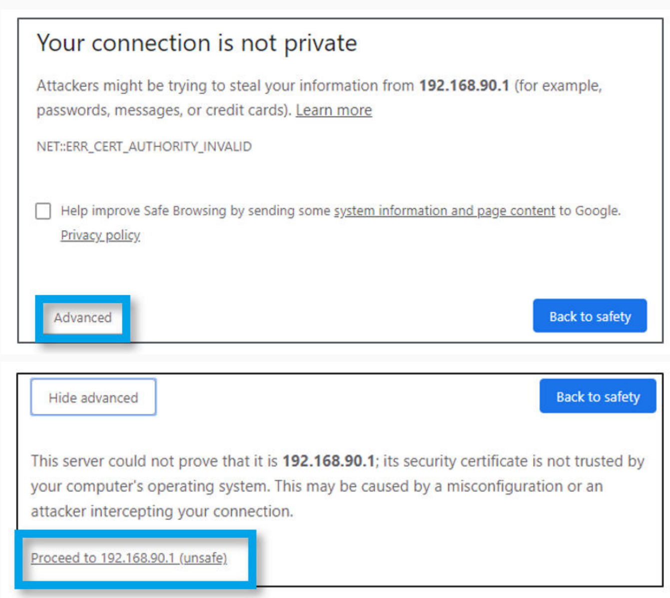
   On Mac:

   - Go to **System Preferences** → **Network** → **Wi-Fi** → Click **Turn Wi-Fi Off**
   - Select the connected Ethernet Port → **Configure IPv4** → Select **Manually** from the dropdown menu → Set your laptop's IP to 192.168.90.251 → Set subnet mask correctly; typical subnet mask is 255.255.255.0

Run the Install Wizard:

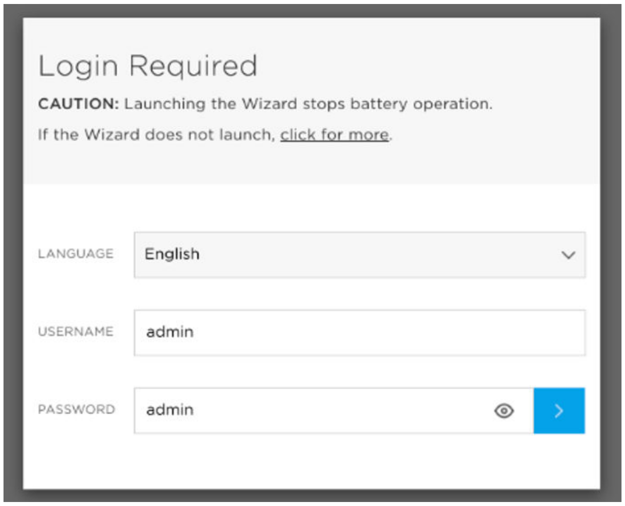
1. Open Google Chrome and enter the following into the address bar: https://192.168.90.1:1111. Press **Enter.**

NOTE: If you see a warning message: Click **Advanced.** then click **Proceed to 192.168.90.1.**



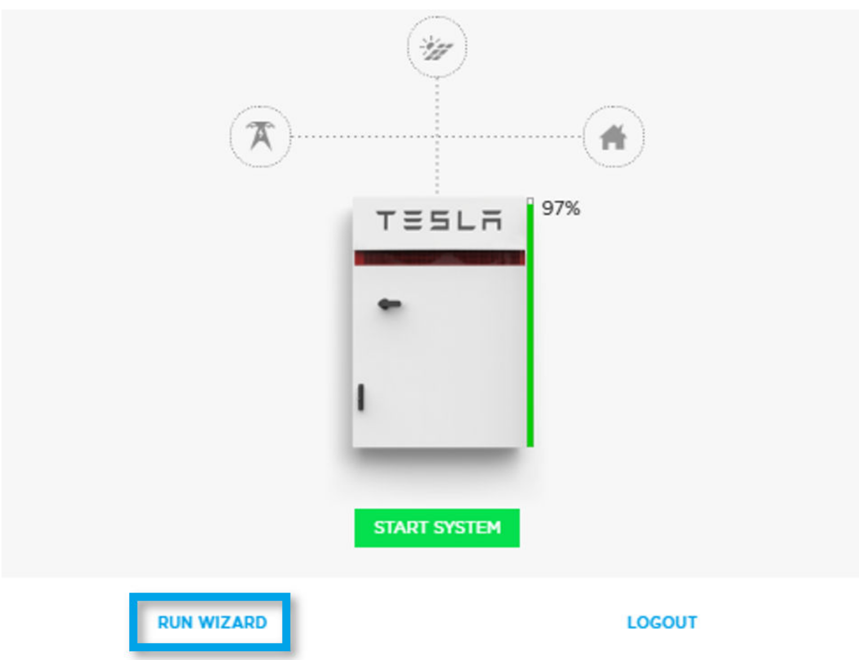
> **NOTE:** If you are unable to access the page, enter the same address but without the final 5 characters, " : 1 1 1 1 ". Then, in the screen that appears, select **Commission System** on the right side of the home page.

2. Log in to the Install Wizard using the following credentials:
   - Username: admin
   - Password:
     - For firmware version 3.31.X and below: admin
     - For firmware version 3.32.X and above: STST-SM-#### (#### is found on the Tesla Site Controller label)



> **NOTE:** The newest versions may have a password printed on the label affixed to the device

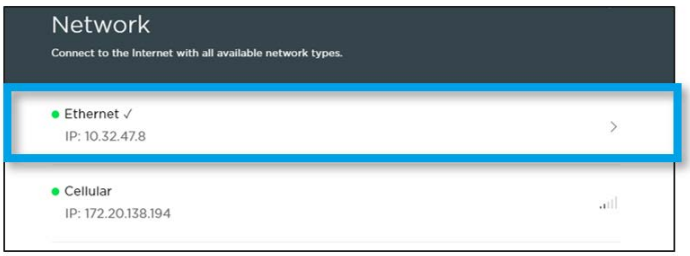
3. Click **RUN WIZARD**:



4. On the Installation Information page, you will enter your site's company name, point of contact's phone, and point of contact's email address. Fill out every field for your site:

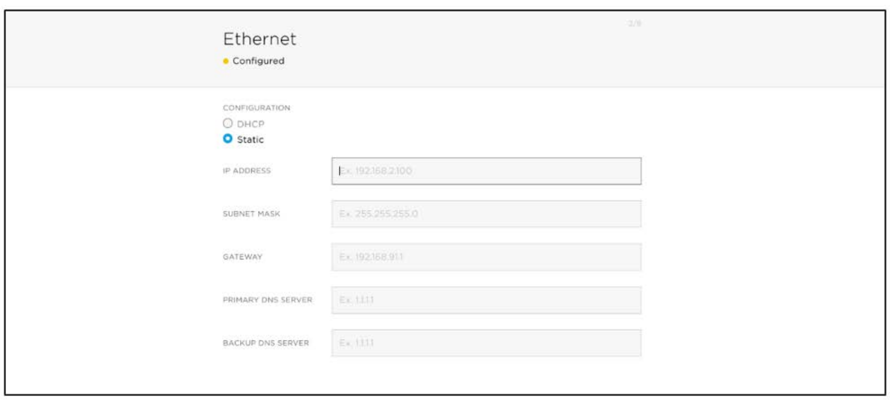
Configure your site's Ethernet connection:

1. On the Network page, select **Ethernet**:



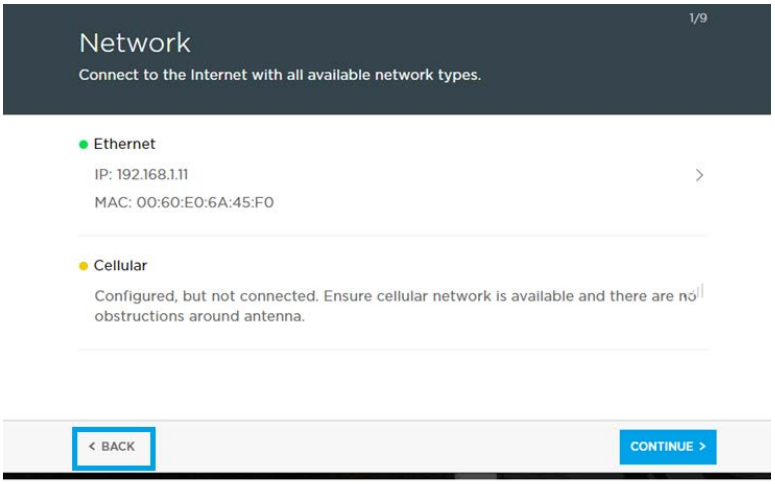
The default network configuration is DHCP. <Connect to the below image starting with> "If your site's Ethernet is Static: Enter all the fields accordingly an click CONNECT."

2. Enter all fields accordingly and click **CONNECT**:

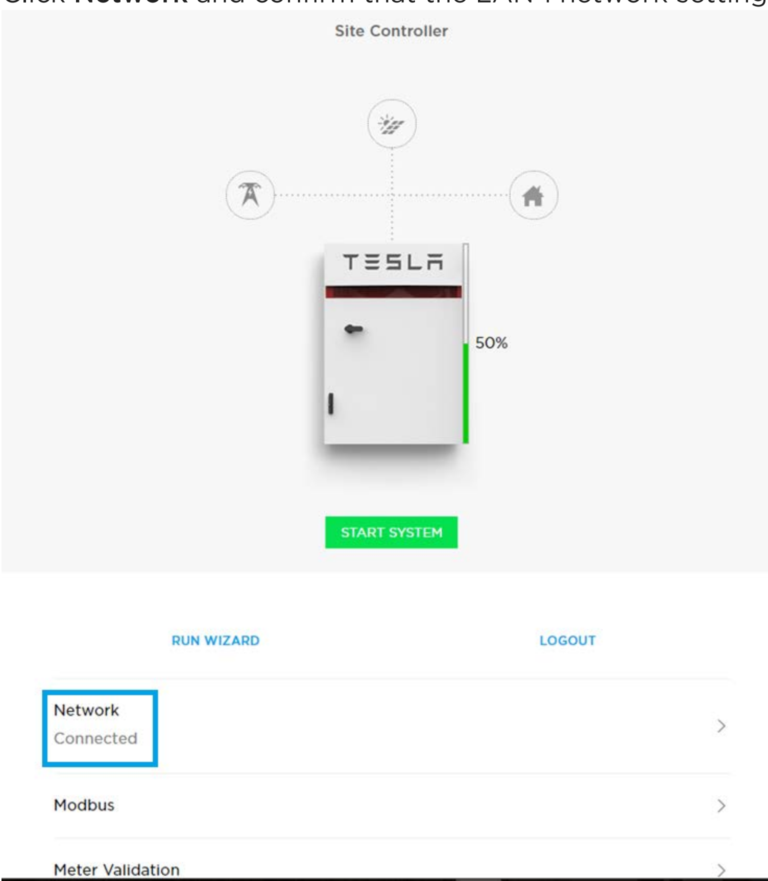


NOTE: These network settings are for the LAN 1 customer connection.

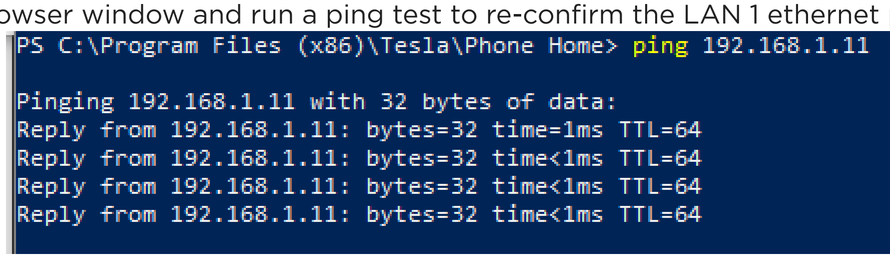
3. Click **Back** at the bottom left corner of the network page:



4. Click **Network** and confirm that the LAN 1 network settings have remained:



5. Close the browser window and run a ping test to re-confirm the LAN 1 ethernet port network settings, for example:



Revision History

Revision #	Date	Description
1.0.0	11-28-2018	• Initial Release
1.0.1	2-22-2019	• Changed all free-form Figure captions and references to inserted captions and references. • Added Table captions. • Standardized Table formats. • Fixed various typos and pagination. • Formatted graphics and Tables in Appendices.
1.1.0	4-16-2019	• General Review and edits • Update SCADA Offering information to reflect current state • Change formatting and figures for standardization and XML conversion
1.2.0	11-18-2019	• Added details on Cellular connectivity parties' scope • Update SCADA Offering information to reflect current state • Update Battery Storage Language • General updates on Field Network Enclosure (FNE) • Network Switch part number updates • Remote connection details update
1.3.0	01-16-2020	• In section 4.2, removed RTAC template reference to Appendix A; removed Appendix A
1.4	August 5, 2020	• Updated names of Powerhub Plus (from Cloud UI) and Powerhub Pro (from Local UI) • Enhanced graphics • Replaced drawings • Added Tesla Site Controller redundancy and Time sync • Added guidance for satellite clocks • Added guidance to Panel PC • Removed RTAC template • Provided guidance on Axiomtek vs Logic • Enhanced FNE and fiber rings per LAN 2 guidance • Updated security firewall section
1.5	August 21, 2020	• Changed product names from Powerhub Pro to Powerhub Local and Powerhub Plus to Powerhub Cloud

Revision #	Date	Description
1.6	September 15, 2020	• Updated Spider switch part number in the *Using Unmanaged Switches* section
1.7	September 29, 2020	• Updated sub-ring guidance – for sites larger than 6 FNEs
1.8	November 11, 2020	• Updated to support gigabit network standard • Revised network switch guidance • Updated Greyhound switch information including part number and sample image • Updated field network enclosure BOM • Updated POI SCADA enclosure standard BOM • Updated I/O section Moxa unit information • Added Panel PC instructions • Removed communication and data flow section • Moved the SCADA interfaces section earlier in the document • Updated architecture diagrams for Gigabit Ethernet standard • Updated Fiber S-Ring architecture diagram
1.9	December 23, 2020	Introduced new name for Standard Site Controller and its enclosure, and for Large Site Controller (computer for sites over 18 MW)
2.0	March 25, 2021	• Extensive changes to clarify and delineate requirements, recommendations, and responsibilities based on Tesla's O&M obligations • Aligned bandwidth requirements • Aligned network switch firmware requirements • Clarified VLAN segregation guidance is for LAN 1 only
2.1	May 24, 2021	• Changes for Tesla Site Controller dual-computer configuration • Removed XFMR RIO from drawings as FNE IO serves this purpose • Modified guidance from *required* to *recommended* for LAN 2 network switch connections • Updated firewall hostnames for logging service (*Through a Customer Firewall* on page 12) • Provided guidance to avoid placing FNEs in direct sunlight (*Field Network Enclosure* on page 30)
2.2	August 11, 2021	• Added the SCADA Acceptance Checklist • Added Tesla admin privileges to jump server requirements (*Alternative Remote Connection Options* on page 14) • Added clarity around data logging traffic via a proxy server through a customer firewall (*Through a Customer Firewall* on page 12) • Adjusted frequency of data logging upload requirement (*Data Logging via Proxy* on page 15)

Revision #	Date	Description
		• Updated the range of IP addresses for RTAC and Tesla Site Controller on LAN 1 and LAN 2 (*Appendix A: Standard IP Address Range* on page 34) • Added requirement that firewall be capable of URL filtering (*Through a Customer Firewall* on page 12) • Added transformer monitoring to *Telemetry* on page 6 • Modified names of Tesla Site Controller components. Now *Standard Tesla Site Controller*, *Standard Tesla Site Controller Enclosure*, and *Large Tesla Site Controller*
2.3	November 11, 2021	• Updated network switch requirements (*Network Switches* on page 25) • Updated fiber cable requirement to single-mode (*Fiber Optic* on page 31)
2.4	March 4, 2022	• Added *Appendix B: Data Proxy Commissioning* on page 35 • Updated the example SCADA readiness spreadsheet • Updated network switch guidance (*Network Switches* on page 25) • Updated Tesla Site Controller redundancy nomenclature (*Network Redundancy* on page 20) • Added *Appendix C: Network Configuration of the Tesla Site Controller's LAN 1 Ethernet Port* on page 38 • Updated SEL RTAC guidance (*Hardware Selection* on page 23)
2.5	April 22, 2022	• *Network Architecture* on page 8 Overhaul • Clarified arbitrage language (*Custom Control Strategies* on page 6) • Standard Division Of Responsibilities update (*Introduction and Scope* on page 4) • Removed SCADA Checklist appendix. See new *Megapack Deployment Checklists* on the Partner Portal. • Tesla's Jump server procedure update/addition (*Alternative Remote Connection Options* on page 14) • Removal/deprecation of Appendix C: Tesla Server Connectivity